

Pre-Minor Progress Report

Surface Ozone Prediction Using a Hybrid Physics-Informed Ensemble Framework

Student	Entry Number
Sai Sindhush Penugonda	2024CS10303
Jayendra Vijay Birhade	2024CS10891

Supervisor: Dr. Ravi Kumar Kunchala Centre for
Atmospheric Sciences (CAS), IIT Delhi
Submission Date: March 8, 2026

Repository: <https://github.com/superionsai/Surface-Ozone-Multi-Model-Forecasting>

1. Project Overview and Objectives

This report documents the pre-execution phase of the ASD390 Minor Area Project, which targets real-time surface ozone (O_3) concentration prediction at the RK Puram CPCB CAAQMS monitoring station in Delhi. Surface ozone is a secondary pollutant—it doesn't come directly from tailpipes or smokestacks. Instead, it forms through a complex photochemical reaction between nitrogen oxides (NO_x) and volatile organic compounds (VOCs) under solar radiation. The problem is that this reaction is heavily nonlinear. Simple regression doesn't work here.

The proposed solution is a hybrid ensemble architecture that blends three complementary modeling paradigms:

- **XGBoost Chemical Kinetics Regression** - handles immediate precursor-to-ozone transformations and regime switching
- **Parameterized LSTM** - captures temporal stochasticity and periodic urban emission cycles (morning rush hour, weekend patterns, seasonal chemistry)
- **Physics-Informed Neural Networks (PINNs)** - enforces atmospheric chemical transport constraints so the model can't hallucinate physically impossible ozone spikes

The ensemble meta-learner (LightGBM) dynamically weights each base model depending on context—trusting the chemical regression during high-UV periods, the LSTM during rush hours, and the PINN as a sanity check whenever base models disagree. Target performance: $R^2 > 0.85$, $MAE < 15 \mu g/m^3$.

2. Pre-Execution Phase - Work Completed

2.1 Literature Review and Architecture Finalization

We studied existing SOTA techniques for atmospheric time-series prediction, including pure CTM approaches, standard ARIMA-based methods, and recent hybrid ML frameworks from literature. Standard time-series models struggle here because they treat the atmosphere as a black box—they see patterns but don't understand chemistry. Pure CTMs, on the other hand, need massive computation and can't deploy in real-time. The proposed architecture bridges both ends: fast enough for operational deployment, physically informed enough to respect the underlying science.

After reviewing this, we finalized the three-model ensemble described above and documented the complete implementation specifications in the attached Technical Paper (Paper 1), including loss function formulations, hyperparameter ranges, and code implementations for each model.

2.2 Data Collection and Preprocessing

Five years of daily CPCB CAAQMS data (2020–2024) were collected from the RK Puram station, Delhi. The raw data arrived as separate yearly CSV files. After merging, standardizing column names, and converting to a proper datetime index, the unified dataset contains **1827 daily observations** across ten pollutant variables: $PM_{2.5}$, PM_{10} , NO, NO_2 , NO_x , NH_3 , SO_2 , CO, O_3 , and Benzene (as VOC proxy).

Missing values—primarily from sensor maintenance and calibration periods—were handled using a multi-stage imputation strategy: time-based linear interpolation for gaps within a reasonable window, followed by forward/backward fill at dataset boundaries. After cleaning, the dataset is complete and ready for model ingestion.

2.3 Feature Engineering

Raw concentrations alone are insufficient for ozone prediction. We engineered the following derived features that capture the critical regime switches in atmospheric chemistry:

Feature	Description	Relevance
VOC/NO _x ratio	Benzene / NO _x	NOx-limited vs VOC-limited regime switch
Leighton ratio	[NO][O ₃] / [NO ₂]	Deviation from photostationary equilibrium
NO ₂ /NO ratio	Photochemical aging indicator	Air mass oxidation state
O3_lag1, NOx_lag1	Previous-day values	Multi-day photochemical accumulation
Rolling means	1-day windows for all species	Smoothed precursor signal

3. Exploratory Data Analysis

3.1 Dataset Statistics

Variable	Mean	Std Dev	Min	Median	Max	Unit
PM2.5	109.0	82.4	7.1	88.0	637.0	µg/m³
PM10	166.1	109.0	16.0	136.0	695.0	µg/m³
NO	14.9	20.5	0.1	7.5	313.0	µg/m³
NO2	40.4	25.2	1.0	37.0	322.0	µg/m³
NOx	50.5	33.2	2.0	43.0	324.0	ppb
NH3	28.5	21.8	0.3	22.0	320.0	µg/m³
SO2	7.0	8.0	0.1	4.4	107.0	µg/m³
CO	1.5	0.9	0.2	1.3	9.2	mg/m³
O3	24.8	16.1	0.4	21.0	103.0	µg/m³
Benzene	6.2	5.6	0.1	4.6	103.0	µg/m³

Table 1. Descriptive statistics for all monitored pollutant variables (RK Puram, 2020–2024).

3.2 Distribution of Pollutant Concentrations

The histogram analysis (Figure 1) reveals that all major pollutants exhibit strongly right-skewed distributions. PM_{2.5} and PM₁₀ show the most pronounced heavy tails, with extreme episodic events-likely dust storms or winter inversions- pushing concentrations well above 500 µg/m³. NO_x species reflect the variability of traffic emissions; under stagnant conditions they spike dramatically. Ozone, by contrast, shows a comparatively more symmetric distribution, which makes sense: its formation is chemically equilibrated rather than directly source-driven.

These non-Gaussian distributions directly justify the use of tree-based and neural models over standard linear regression-models that assume normality will simply fail on this dataset.

Distribution of Pollutant Concentrations

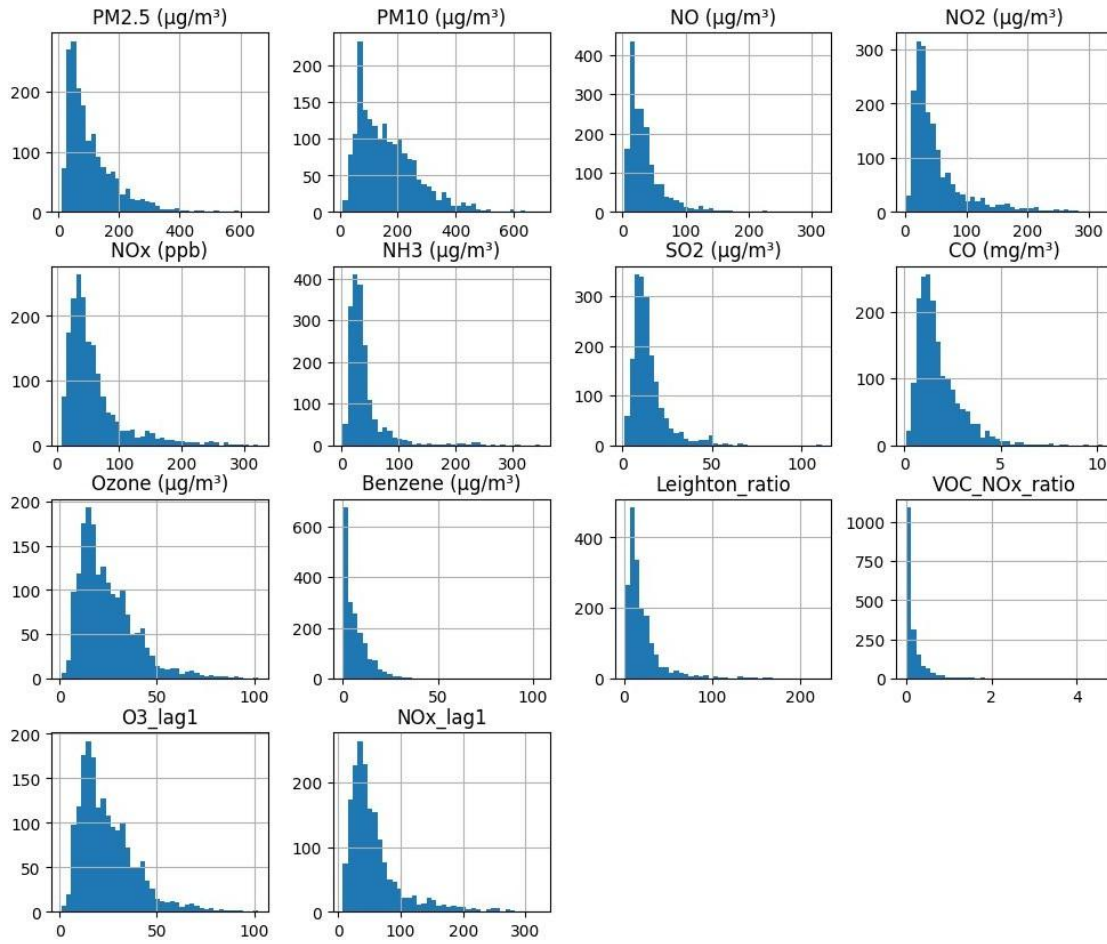


Figure 1. Distribution of all pollutant concentrations over the 2020–2024 observation period (RK Puram station). All variables exhibit right-skewed distributions with significant heavy tails.

3.3 Photochemical Equilibrium: Leighton Ratio Analysis

The Leighton ratio is defined as $\Phi = [\text{NO}][\text{O}_3] / [\text{NO}_2]$, and it serves as a direct diagnostic for whether the atmosphere is in photostationary equilibrium. Under clean conditions, this ratio hovers near a predictable equilibrium value. Any significant deviation indicates that additional chemical pathways—primarily VOC-driven peroxy radical chemistry—are active.

As shown in Figure 2, the Leighton ratio fluctuates wildly across the observation period, with values ranging from near zero to above 200. This is not equilibrium behavior. It confirms that the RK Puram atmosphere is frequently VOC-perturbed, meaning that free radical chemistry is actively pushing ozone formation beyond what the simple NO–NO₂–O₃ photocycle alone would predict. This is exactly the kind of environment our PINN model is designed for.

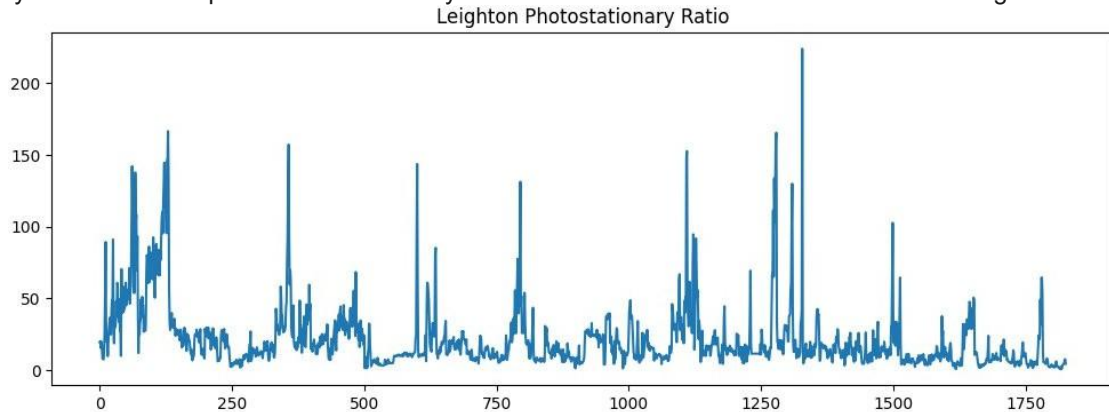


Figure 2. Leighton photostationary ratio over the full 1827-day observation period. Large deviations from equilibrium (values $\gg 1$) confirm

3.4 Ozone vs. Leighton Ratio - Similarity Analysis

Overlaying ozone concentration against the Leighton ratio (Figure 3) reveals a striking structural similarity between the two time series. During high-ozone episodes, the Leighton ratio spikes in tandem, confirming that the ratio captures the same underlying photochemical signal driving ozone buildup. This validates using the Leighton ratio as a physics-grounded feature in our model rather than just relying on raw precursor concentrations.

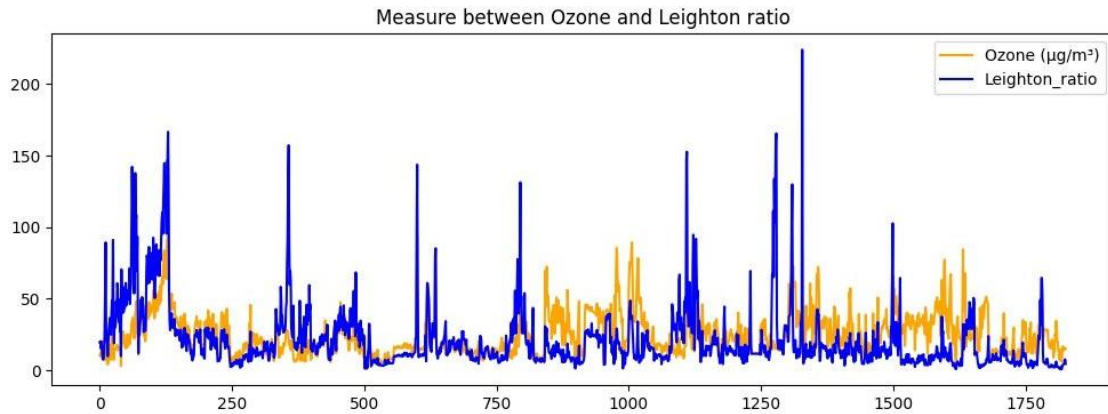


Figure 3. Comparison of observed ozone (orange) and Leighton ratio (blue) over the full observation period. Peaks align closely, confirming that the Leighton ratio tracks photochemical ozone production dynamics.

3.5 Pollutant–Ozone Relationship Analysis

To understand how each precursor group relates to ozone, we sorted all observations by ozone concentration and plotted coincident pollutant levels. The results are broken into four categories below (Figures 4–7)

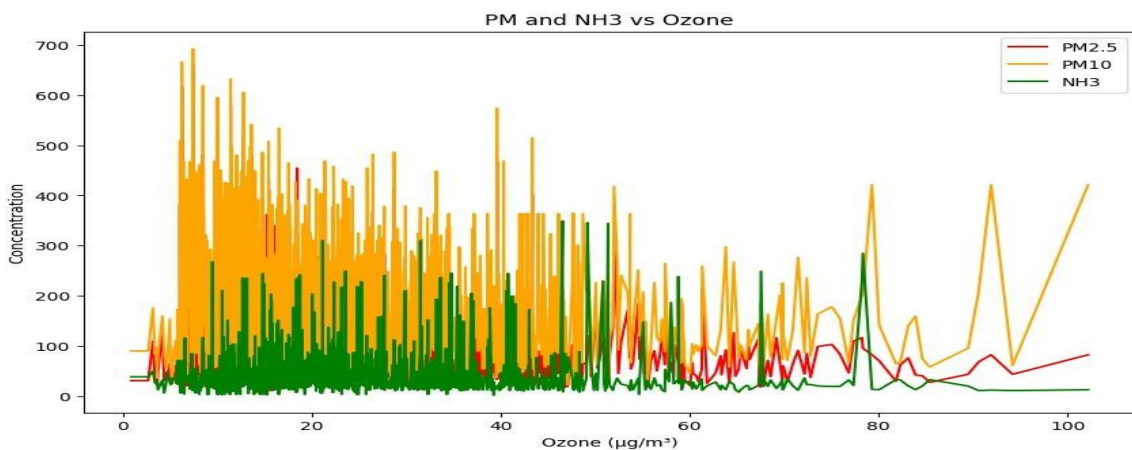


Figure 4. $PM_{2.5}$, PM_{10} and NH_3 vs Ozone.

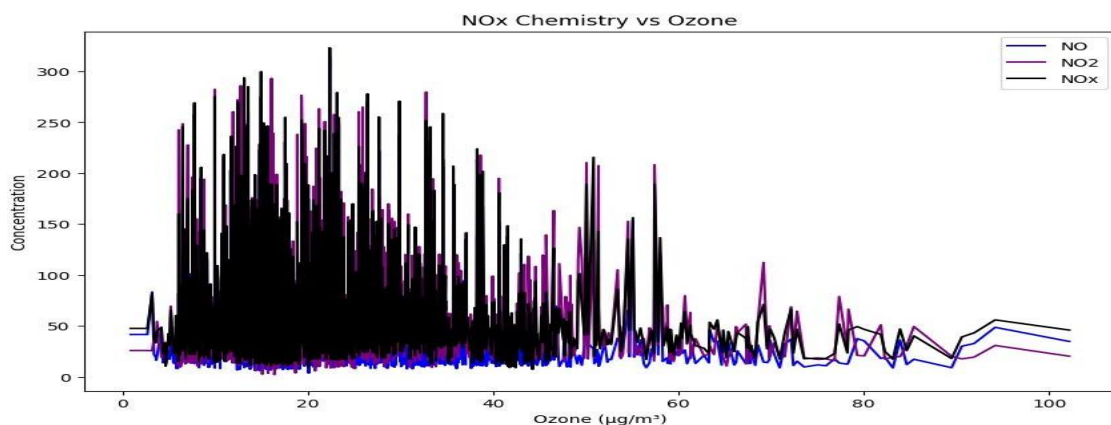


Figure 5. NO , NO_2 , NO_x Chemistry vs Ozone.

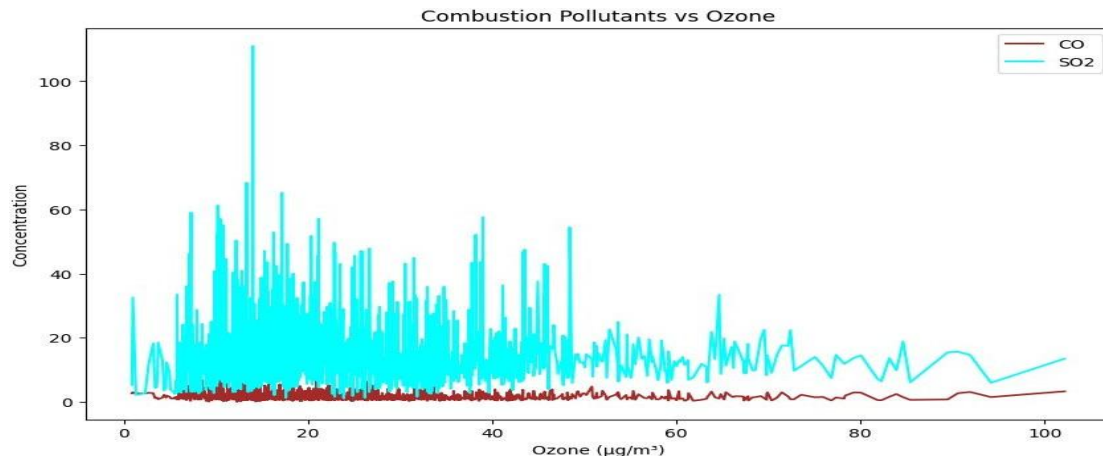


Figure 6. Combustion pollutants (CO, SO₂) vs Ozone.

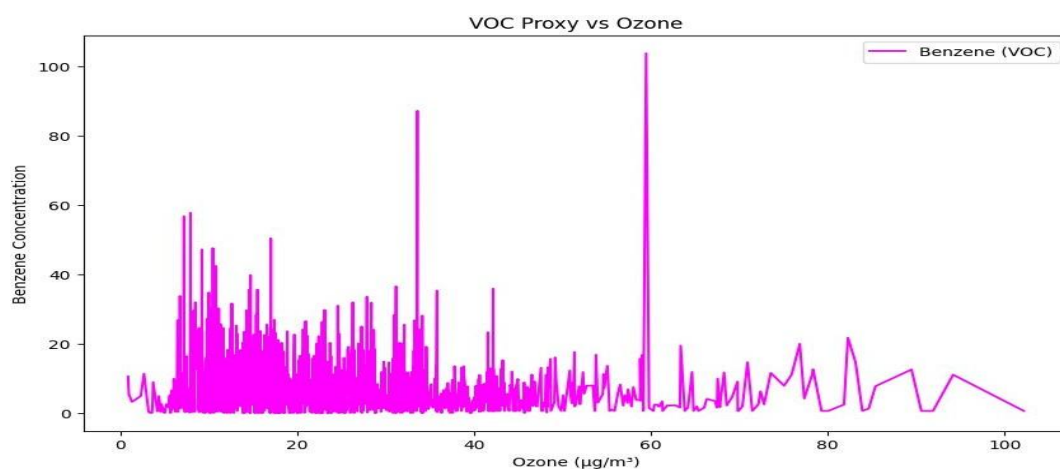


Figure 7. Benzene (VOC proxy) vs Ozone.

Key findings from these plots:

- NO_x species (NO, NO₂, NO_x) show a clear inverse relationship with ozone - this is the classical NO titration effect, where NO reacts with O₃ to regenerate NO₂. High ozone only accumulates when NO has been depleted. We also observe that the nitrogenous species co-exist with similar ratio w.r.t Ozone.
- PM_{2.5} and PM₁₀ show no strong direct correlation with ozone levels, confirming that particulate and photochemical pollution are mostly independent formation pathways at this station.
- CO and SO₂ show moderate but noisy variability without a clear directional trend - CO serves primarily as a VOC proxy in our model rather than a direct ozone precursor.
- Benzene (VOC proxy) shows episodic spikes that partially co-occur with elevated ozone, confirming its role in driving peroxy radical chemistry and VOC-limited ozone formation.

3.6 VOC–NO_x Chemical Regime Analysis

The VOC/NO_x ratio determines whether ozone formation is constrained by VOC availability or by NO_x concentrations- this is one of the most important diagnostics in ozone prediction. As shown in Figure 8, the ratio remains below 1.0 for the vast majority of the observation period, with occasional spikes near 4.5 during episodic high-VOC events. This indicates that RK Puram operates predominantly in a **VOC-limited regime**-consistent with a high-traffic urban environment where NO_x from vehicles is abundant, but reactive VOC emissions are comparatively lower.

The implication for control: reducing VOC emissions at this location should be more effective for ozone reduction than cutting NO_x alone. It also means our model needs to be sensitive to benzene-proxy fluctuations, which is why we retain it explicitly as a feature even though it's noisy.

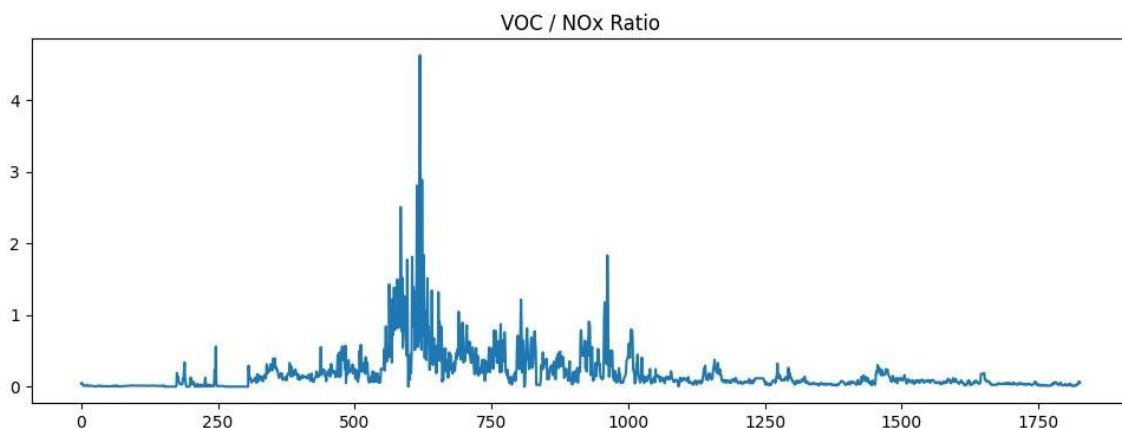


Figure 8. VOC/NO_x ratio over the full observation period. Ratio predominantly below 1.0 confirms a VOC-limited photochemical regime at RK Puram.

4. Finalized Model Architecture

Based on all the data analysis above, the model selection is justified as follows and finalized:

Model	Role	Justification from Data Analysis
XGBoost Regression	Chemical kinetics + regime switching	Non-Gaussian skewed distributions; NO _x -limited vs VOC-limited regime switches require piecewise nonlinear boundaries
Parameterized LSTM	Temporal stochasticity + periodic cycles	Lagged NO _x correlation (-0.10) shows memory effects; emission patterns repeat daily/weekly; LSTM gating handles this natively
Unsupervised PINN	Physical constraint enforcement	Leighton ratio deviations confirm equilibrium violations; ADR equation must be enforced to prevent hallucinated predictions
LightGBM Meta-Learner	Dynamic ensemble weighting	Each base model has orthogonal error modes; meta-learner routes trust based on UV, temperature, hour-of-day context

Table 2. Finalized model components and their data-driven justification.

4.1 XGBoost - Chemical Kinetics Regression

XGBoost builds an ensemble of decision trees sequentially, using second-order Taylor expansion of the loss function (incorporating the Hessian). This lets it efficiently learn piecewise nonlinear relationships—exactly what’s needed when the chemistry is switching between VOC-limited and NO_x-limited regimes. SHAP-based interpretability is included to track which precursors are driving each prediction.

4.2 LSTM - Temporal Pattern Learning

A stacked bidirectional LSTM (128 → 64 units, dropout 0.3) processes sequences of length 24 (6 hours at 15-min resolution). The bidirectional design means the model sees both past and recent context simultaneously. Huber loss is used during training for robustness against outlier spikes. A Temporal Fusion Transformer (TFT) variant is also available if the standard LSTM underfits.

4.3 PINN - Physics Constraint Enforcement

The PINN modifies the standard loss function to include a physics residual term:

$$L_{\text{total}} = L_{\text{data}} + \lambda \cdot L_{\text{physics}}$$
where $L_{\text{physics}} = \text{MSE}(\partial \mathbf{C} / \partial t - R(\mathbf{C}, [\text{NO}_x], [\text{VOCs}]))$, and R is derived from the simplified advection-diffusion-reaction equation at a point monitoring station (no spatial gradients). $\lambda \in [0.1, 1.0]$ controls the physics-data trade-off. Training is unsupervised with respect to the physics loss term-the network learns to respect the reaction equations without explicit labeling.

5. Timeline and Next Steps

Phase	Tasks	Status	Target
Pre-Execution (Completed)	Literature review, architecture design, data collection, EDA, feature engineering	✓ Done	March 3, 2026
Abstract Model Development	XGBoost training + SHAP analysis, LSTM sequence modeling, PINN training loop	✓ Done	March 8, 2026
Modification and Ensemble Integration	LightGBM meta-learner, stacking pipeline, uncertainty quantification	In progress	By E.O.M
Evaluation & Validation	TimeSeriesSplit CV (5 folds), $R^2 > 0.85$ target, Peak ozone episode accuracy	Planned	By April 2026

Table 3. Project timeline and phase completion status.

We expect the full model to be complete in approximately one month from this submission. After that, the remaining project period will be used for evaluation, refinement, and write-up. The complete codebase including all preprocessing scripts, model implementations, and training notebooks is maintained in the GitHub repository linked above.

6. Summary and Conclusion

The pre-execution phase of this project is fully complete. All the analysis and data collection tasks are done with relevant plots and theoretical studies to understand how the model should behave on this specific station's data. Key conclusions:

- The RK Puram dataset is of sufficient size (1827 observations) and diversity to train a robust multi-model ensemble
- Non-Gaussian pollutant distributions confirm that tree-based and neural models are appropriate; standard regression would fail
- The Leighton ratio analysis confirms active VOC-peroxy radical chemistry, validating the need for a PINN constraint layer
- The VOC-limited regime diagnosis means benzene proxy is a critical feature-the model must be sensitive to its episodic spikes
- The ozone–NOx inverse relationship confirms classical titration dynamics are well-represented in this dataset
- All derived features (Leighton ratio, VOC/NOx, lag variables) have been computed and validated against expected atmospheric behavior

The project sees good light going forward. The data analysis confirms that our station aligns well with the conditions under which the proposed ensemble architecture is expected to work best. We are confident in hitting the >85% accuracy target with proper hyperparameter tuning and ensemble integration.